

Course Code: CSE254	Course Title: Microprocessor and Microcontroller Laboratory	L-T-P-C	0 -0	2	1
	Type of Course: Laboratory Only				
Version No.	2.0				
Course Pre-requisites	NIL				
Anti-requisites	NIL				
Course Description	This course introduces the assembly level language programming of 8086. The course introduces the core concept of microprocessor and develops in students the assembly language programming skills along with real time applications of microprocessor. It gives a practical training to students to perform interfacing peripheral devices with 8086 microprocessors. This lab focusses mainly on software and few interfacing programs with microprocessor				
Course Objective	The objective of the course is to familiarize the learners with the concepts of Microprocessor and Microcontroller Laboratory and attain SKILL DEVELOPMENT through EXPERIENTIAL LEARNING techniques.				
Course Outcome	After successful completion of course, students shall be able to (i) Learn 80x86 instruction sets and gain the knowledge on how assembly language works. (ii) Implement programs written in 80x86 assembly language. (iii) Explore functioning of hardware devices and interfacing them to x86 family. (iv) Implement basic 8051 microcontroller programs.				
Course Content:					
	:	Write an Assembly Language Program (ALP) to perform Arithmetic operations like Addition, subtraction, Multiplication and Division on two numbers			
	:	Write an ALP to add two Binary Coded Decimal (BCD) numbers			
	:	Write an ALP To move 8-bit contents of array from one memory location to another memory location			
	:	Write an ALP to find the sum of N consecutive numbers			
	:	Write an ALP to sort N numbers in ascending/descending order using Bubble sort technique Write an ALP to print N Fibonacci numbers.			

	:	Write an ALP to search a key element in a list of numbers using linear search
	:	Write an ALP to read the current time from the system and display on screen Write an ALP to check whether a string is Palindrome or not
	:	Write an ALP to search a key element in a list of numbers using binary search
	:	Write an ALP to read the current date from the system and display on screen
	:	Write an ALP to read two strings from the keyboard and check whether they are equal or not.
8255 Interfacing Experiments		
	:	Design and develop an ALP to drive a Stepper Motor interface and rotate the rotor in specified direction (clockwise or anti-clockwise) by N steps
	:	Design and develop an ALP program using Logic Controller to multiply (X*Y)
8051 Microcontroller Experiments		
	:	Design and develop 8051 ALP program to store values in registers and swap the contents of Registers
	:	Design and develop 8051 ALP program to perform arithmetic operations
	:	Design and develop 8051 ALP program to perform FIBONACCI series
	:	Design and develop an 8051 ALP to drive a Stepper Motor interface and rotate the rotor in specified direction (clockwise or anti-clockwise) by N steps

Targeted Application & Tools that can be used: MASM,

Professionally used software - KEIL software

Text Book

Douglas V Hall SSSP Rao, "Microprocessor and Interfacing", 3rd edition, Mc Graw Hill , Higer Education, 2012.

Barry B Brey, " The Intel Microprocessors", 8th edition, Pearson , 2014.

References

Muhammad Ali Mazidi, Janice Gillispie Mazidi, Danny Causey, "The x86 PC Assembly Language Design and Interfacing", 5th Edition, Pearson, 2013.

Muhammad Ali Mazidi, "Microprocessors and Microcontrollers", First Impression, Pearson Education.

<https://nptel.ac.in/courses/108105102>

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